Lifting Bot Challenge

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November 4, 2023

PURPOSE

The objective was to create a vex robot to collect and lift foam balls into various bins. There are three bins, with ascending points for the bins farther off the ground. The vex robots in the class would then compete with one another under a time limit, and the highest point total would win.

INVESTIGATE

The vex bot must meet various *needs* to be competitive. First, it must have the proper ability to collect the balls, whether it be by a claw mechanism or a type of shovel. The bot needs to have a fast and efficient lifting mechanism to carry the balls to a higher height. The bot also needs to be wider than the lift bot. Lastly, the bot needs to be easily controllable to maximize performance efficiency. Ideally, a maximization of these three attributes will allow the bot to put all the foam balls into the top bin within the time limit.

What materials were used?

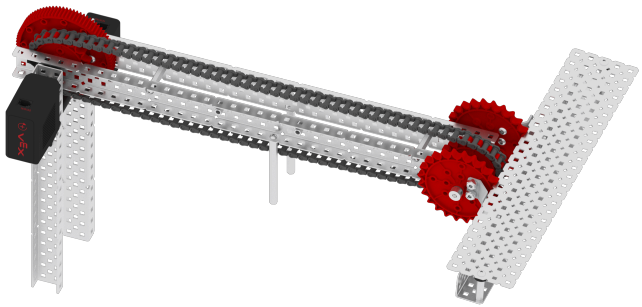
Qt.	Material	Qt.	Material
4	4 U-Channel	20	Screw Holders
4	5 U-Channel	7	Metal Rods
6	L- Bracket	2	Red Gears
4	Wheels	4	Plastic Pinions
6	Motors	2	3D printed plastic sheets
30	8-32 Nut	8	Wire connectors
30	8-32 x 0.375 in screw	1	Radio
30	8-32 x 0.5 in screw	1	Battery
20	0.125 in to 0.875 in spacers	1	Robot Brain + Controller

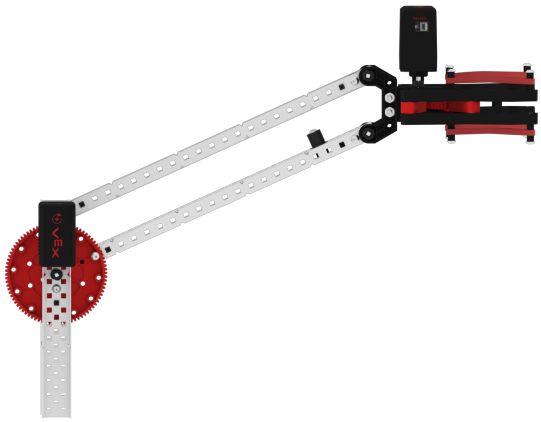
RESEARCH

For the initial research, we looked at different videos in the D2L to identify which of the lift types would suit our purposes. The scissor lift mechanism and double lift seemed

too slow and heavy. The other mechanisms seemed to be slow, hard to control, or inefficient. One method considered was the chain-bar arm (featured below to the left). However, the chain mechanism seemed to create unnecessary friction and did not align with our intended goal.

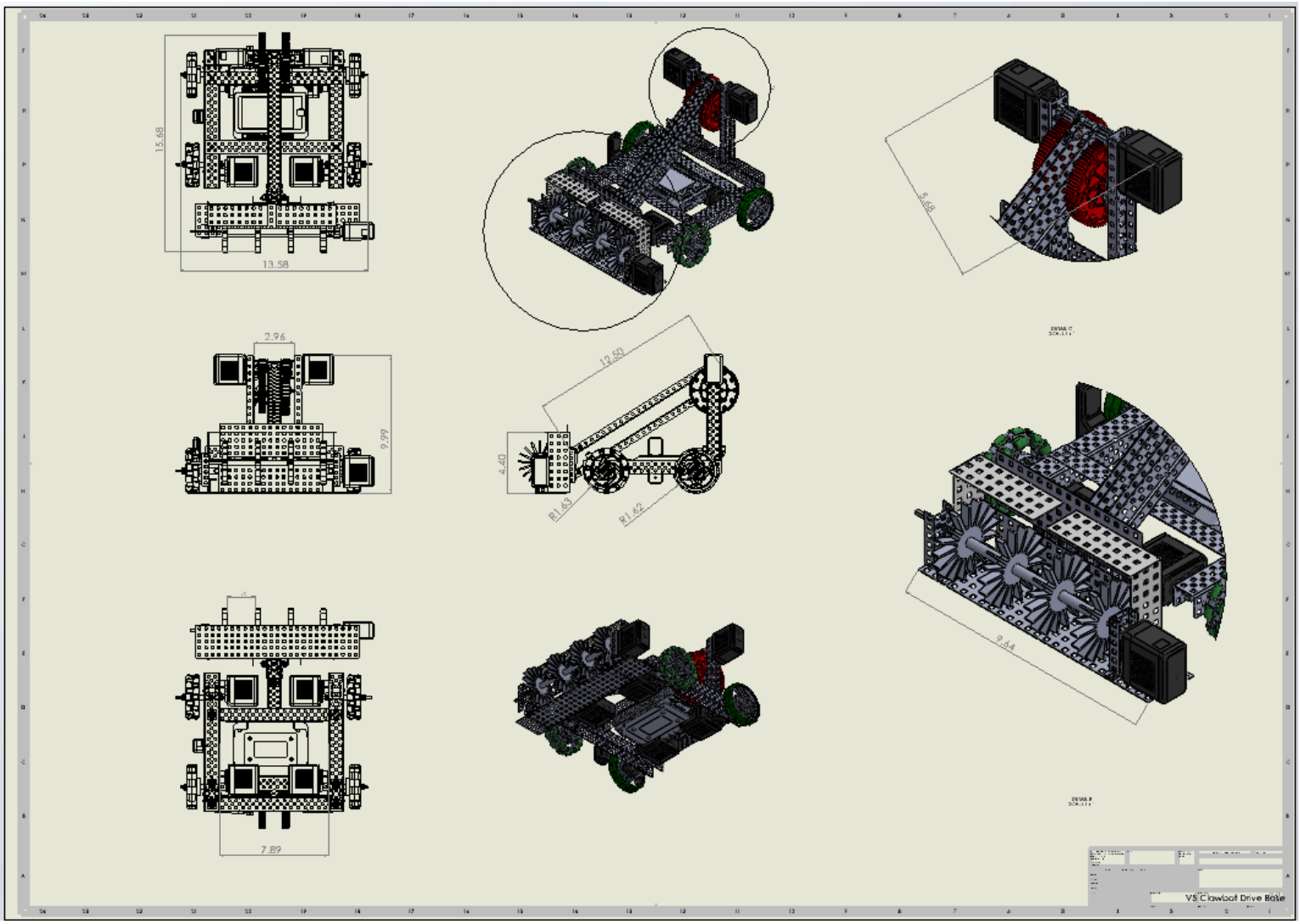
The bar lift (featured below to the right) was ultimately decided upon due to its stability, quick adjustment, and lack of unnecessary weight. We followed the instructions in the sample arm kit, which can be accessed at [276-6009-750-Rev5.pdf](276-6009-750-Rev5.pdf).





Then, we researched various intake mechanisms, including the claw and the "loader" scoop. However, we ultimately decided upon a mechanism based on a model that was featured in class, using motors to intake the foam balls and hold them in place. Then, the foam balls could be released either by reversing the direction or by releasing them over the top (for higher blocks).

PLAN AND DESIGN



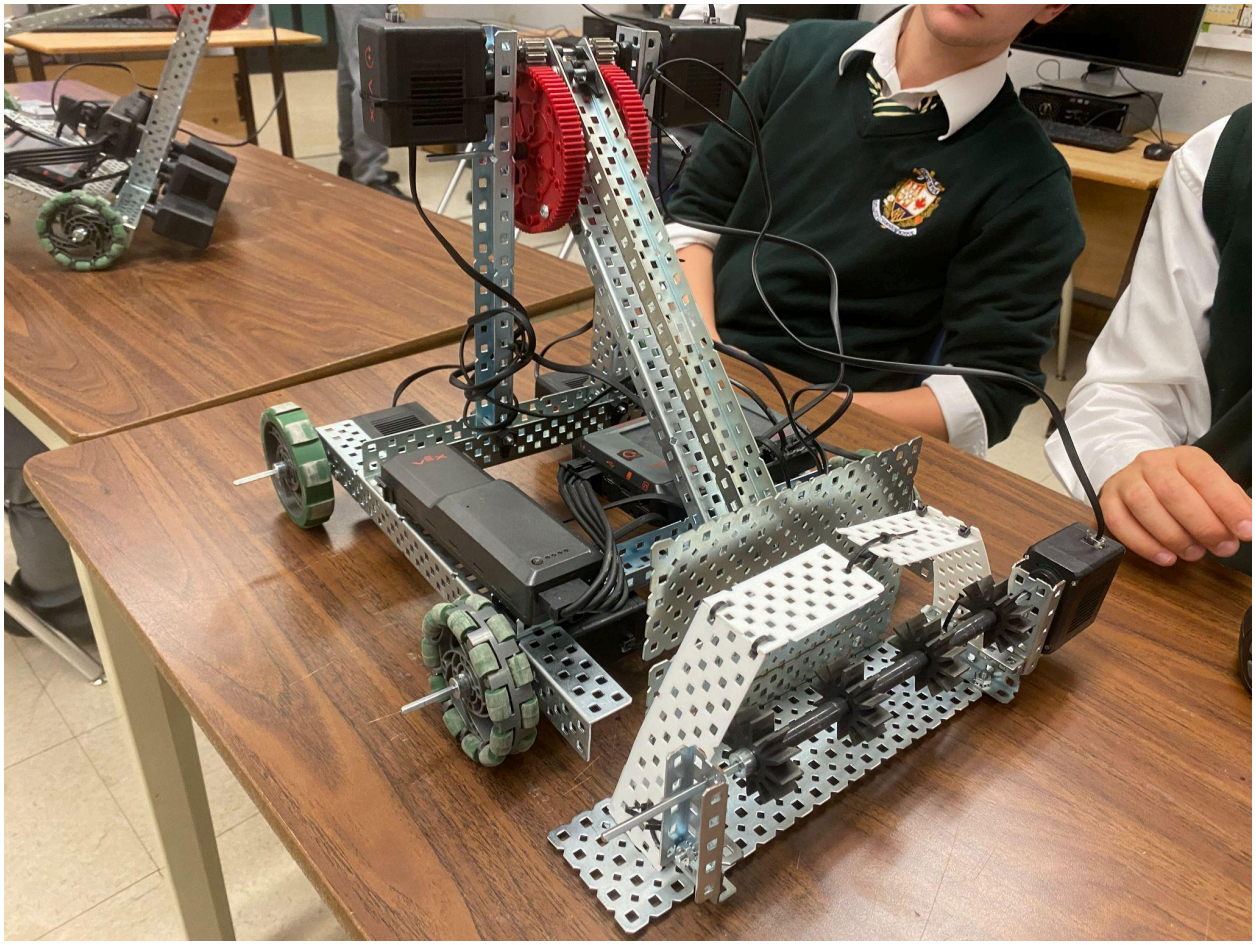
The following are designs created using CAD of what the bot should look like.

The plan was to:

1. Acquire the materials and components to build the car.
2. Reassemble the base of the robot, making it larger and switching the wheels.
3. Assemble the stand for the arm and the arm itself.
4. Build the motor intake mechanism.
5. Test and modify the model where necessary.
6. Perform the final test with the foam balls.

CREATE

Following the previous instructions, we assembled the vex lift bot. **Here is an image:**



What problems did the car have?

Initially, the motor for the lifting mechanism overheated, which was corrected by optimizing both motors to lift and lower the intake.
Then, the foam balls would fall out of the sides of the intake mechanism when the car accelerated. To adjust for this, plastic sheets were added to the sides of the mechanism to enclose the balls with a limited weight increase.

EVALUATE AND REFLECT

Test

The car was tested by competing…

Performance: ***32 points; 16 Balls in the 2-point bin***

Successes

Our team designed an effective lift bot that collected the most points during the competition. The base of the bot was stable and did not tip over. The arm of the bot was tall enough to comfortably lift the mechanism into the second bit yet light enough that the bot did not tip over. Also, the intake mechanism was fairly successful at intaking 2-3 foam balls during the trial and releasing them into the bin. Notably, the plastic sidings near the intake were effective at keeping the balls from falling over the sides and back.

What would you we differently

For future iterations, there are several aspects of the design that could be adjusted to improve the performance. Firstly, there were two instances during the run where the foam balls were stuck between the plastic pinions in the intake mechanism. The pinions can be placed closer together with fewer spacers to reduce the likelihood of future occurrences. Secondly, to reduce the strain on the lifting mechanism and motors, we would use lighter plastic plates for the intake instead of heavy metal counterparts. Thirdly, a future design would optimize the intake mechanism to intake balls through the front and release them over the top without getting stuck. This would allow the vex bot to dispense the balls at a greater height and without having to switch the direction of the motors.

What we have learned

We learned that multiple intake mechanisms have a speed-to-risk ratio when it comes to the ability to pick up balls. For example, a motor on the head slows the speed to pick up the balls, but it is easier to gather. This, compared to a flat scoop, has the fastest pickup time but is hardest to control and dispense. Through this analysis, we determined that the optimal head would utilize a scoop with a moving joint between the arm and head that allows for speed to pick up balls and further control of the storage.

What advice would we give someone about to start the same project?

A group can yield an excellent score using the second bin rather than the third. Reaching the third bin often requires more precision and a longer lifting mechanism, usually meaning a smaller intake mechanism to prevent the bot from tipping over. The second bin can be reached faster and uses a much larger lifting mechanism, meaning more balls per trip to the bin. Thus, be strategic about which bin the group is intending on using and specialize the lift bot to be *the best* at that specific height. Make your bot quick, easy to control, and capable of carrying many foam balls securely. From what we observed in the trials, we aim to be the group that manages to put 5 balls in the second bin quickly, over the group that puts 2 or 3 in the top bin.

Resources

Vex model designed by Michael Mohn

VEX Robotics V5 Clawbot | 3D CAD Model Library | GrabCAD